Chapter 4

LaGrange's Theorem

By the **order** of a finite group **G**, written $\circ(\mathbf{G})$, we mean the number of elements in **G**. In this chapter we will see how numerical properties of $\circ(\mathbf{G})$ imply algebraic properties of the group **G** itself.

Theorem 31 tells us that a cyclic group of order k consists of the first k powers of the generator, where the operation is done by adding the powers modulo k. Thus all finite cyclic groups of the same order have virtually identical tables. Theorem 16 tells us that all groups of order 3 are cyclic, and therefore look alike. Clearly the same is true for groups of order 1 or 2. On the other hand, $\mathbf{K}_4$ is a non-cyclic group of order 4. These observations raise the following question.

Question *For which positive integers n is it true that every group of order n is cyclic?*

The smallest value of n for which we do not yet have an answer is $n = 5$.

Theorem 36. *Every group* **G** *of order 5 is cyclic, and therefore looks exactly like* $\mathbf{Z}_5$.

Proof: (Outline) Let b be any element of **G** other than the identity. Explain why $\circ(b)$ must be either 2,3,4 or 5.

(i) Show that if $\circ(b) = 4$, then **G** would have at least 8 elements. [Fill in the first 4 rows of the table listing the elements as e, b, b^2, b^3. Then add a 5th column for a new element c and explain why $c,\ d := b * c,\ f := b^2 * c,\ g := b^3 * c$ would be 4 distinct new elements.]

(ii) Show that if $\circ(b) = 3$, then **G** would have at least 6 elements. [Fill in the first 3 rows of the table listing the elements as e, b, b^2. Then add a 4th column for a new element c and explain why $c,\ d := b * c,\ f := b^2 * c$ would be 3 distinct new elements.]

(iii) Show that if $\circ(b) = 2$, then **G** would have at least 6 elements. [Fill in the first 2 rows of the table listing the elements as e, b. Then add a 3rd column for a new element c and explain why $c,\ d := b * c$ would be 2 distinct new elements. Add columns for c and d and fill in the 2 columns below them. Now let f be the 5th element and explain why $f,\ g := b * f$ would again be 2 distinct new elements.]

Theorem 37. *Every group* **G** *of order 7 is cyclic, and therefore looks exactly like* $\mathbf{Z}_7$.

Proof: (Outline) Follow the strategy of Theorem 36, adding the necessary additional cases.

What about groups of order 9? Here is a helpful observation. The Cartesian plane $\mathbf{R}^2$, viewed as an additive group, is obtained by taking ordered pairs (a,b) from the additive group of real numbers $\mathbf{R}$ and defining the operation of $\mathbf{R}^2$ by applying the operation of $\mathbf{R}$ in each coordinate:

$$(a_1,b_1)+(a_2,b_2) := (a_1+a_2,\ b_1+b_2).$$

The same idea works with any pair of groups $\mathbf{H} = \langle H; *_H \rangle$ and $\mathbf{K} = \langle K; *_K \rangle$. We define the **Cartesian product** of $\mathbf{H}$ and $\mathbf{K}$ to be the group

$$\mathbf{H} \times \mathbf{K} = \langle H \times K; * \rangle$$

whose elements are ordered pairs (h,k), where $h \in H$ and $k \in K$, and whose operation is done by applying the operations of $\mathbf{H}$ and $\mathbf{K}$ in each coordinate:

$$(h_1,k_1) * (h_2,k_2) = (h_1 *_H h_2,\ k_1 *_K k_2).$$

Theorem 38. *If* $\mathbf{H}$ *and* $\mathbf{K}$ *are groups, then* $\mathbf{H} \times \mathbf{K}$ *is a group. If they are both finite, then* $\mathbf{H} \times \mathbf{K}$ *is finite and* $\circ(\mathbf{H} \times \mathbf{K}) = \circ(\mathbf{H}) \cdot \circ(\mathbf{K})$.

Problem 39. *Use the Cartesian product to construct a group of order 9 that is not cyclic.*

Problem 40. *For exactly which of the values* $n = 1,2,3,4,5,6,7,8,9$ *is it true that every group of order n is cyclic? Make a conjecture as to when in general it is true that every group of order n is cyclic.*

We will settle this question by proving a more general theorem about finite groups due to J. L. LaGrange (1736-1813). First we need some more empirical data. Use Theorem 28 to find all of the subgroups of the four groups in the table below, and list each of their orders. What does this table suggest about the relationship between the order of a finite finite group and the orders of its subgroups?

$\mathbf{G}$	$\circ(\mathbf{G})$	$\circ(\mathbf{H})$ for subgroups $\mathbf{H}$ of $\mathbf{G}$
$\mathbf{S}_3$		
$\mathbf{Z}_{12}$		
$\mathbf{P}_{abc}$		
$\mathbf{K}_4$		

To see why this relationship might hold in general, consider the subgroup $\mathbf{H} := \{0,4,8\}$ of $\mathbf{Z}_{12}$. We can generate new elements of $\mathbf{Z}_{12}$ from $\mathbf{H}$ by applying the operation. If we start with $1 \notin \mathbf{H}$, then $\{0+1,4+1,8+1\} = \{1,5,9\}$ gives us 3 new elements of $\mathbf{Z}_{12}$. Choosing $2 \notin \{0,4,8,1,5,9\}$, we obtain 3 more new elements $\{0+2,4+2,8+2\} = \{2,6,10\}$. Doing the same thing with $3 \notin \{1,4,8,1,5,9,2,6,10\}$ will give us the last 3 elements of $\mathbf{Z}_{12}$. What we find is that $\mathbf{Z}_{12}$ is the disjoint union of 4 sets, each having the same number of elements as $\mathbf{H}$:

$$\mathbf{Z}_{12} = \{0,4,8\} \cup \{1,5,9\} \cup \{2,6,10\} \cup \{3,7,11\}.$$

Thus $\circ(\mathbf{Z}_{12}) = 4(\circ(\mathbf{H}))$ and therefore $\circ(\mathbf{H}) | \circ(\mathbf{Z}_{12})$.

In general we say that a collection $\mathscr{P}$ of subsets of a set A is a **partition** of A if every element of A is in exactly one of these sets, that is,

- each element of A is in some set of $\mathscr{P}$ (in symbols, $\bigcup \mathscr{P} = A$) and

- two different sets X and Y in $\mathscr{P}$ have empty intersection. (We say X and Y are **disjoint**.)

We will now see that every subgroup of every group produces a partition like the one we have seen in $\mathbf{Z}_{12}$.

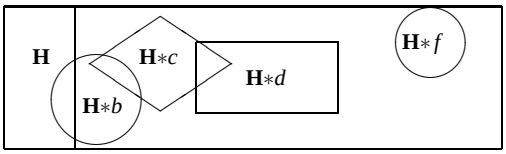
Let $\mathbf{H}$ be a subgroup of a group $\mathbf{G}$. For each element $b \in \mathbf{G}$ we define the **right coset** generated by b to be the set

$$\mathbf{H} * b := \{h * b \mid h \in \mathbf{H}\}.$$

$+$	0	4	8	1	5	9	2	6	10	3	7	11
0	0	4	8	1	5	9	2	6	10	3	7	11
4	4	8	0	5	9	1	6	10	2	7	11	3
8	8	0	4	9	1	5	10	2	6	11	3	7

In the above example, the right cosets of $\mathbf{H} := \{0,4,8\}$ form a partition of $\mathbf{Z}_{12}$. The right coset generated by each element occurs immediately below it in the table for $\mathbf{Z}_{12}$.

We would like to know if the collection $\mathscr{P}$ of right cosets of an arbitrary subgroup $\mathbf{H}$ of an arbitrary group $\mathbf{G}$ form a partition like the one we have seen in $\mathbf{Z}_{12}$. At the moment, all we know about the right cosets of $\mathbf{H}$ is that they are a collection of subsets of $\mathbf{G}$. One coset is $\mathbf{H} = \mathbf{H} * e$ itself. This might be illustrated as follows.



The next two lemmas tell us that this picture is not very accurate.

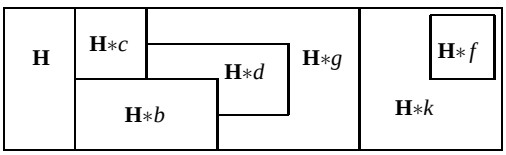
Lemma 41. *If $b \in \mathbf{G}$, then b is in the right coset $\mathbf{H} * b$ that it generates. In particular, every element of $\mathbf{G}$ is in some right coset.*

Lemma 42. *Let $b, c \in \mathbf{G}$.*

(i) *If $c \in \mathbf{H} * b$, then $\mathbf{H} * c = \mathbf{H} * b$.*

(ii) *If $c \notin \mathbf{H} * b$, then $(\mathbf{H} * c) \cap (\mathbf{H} * b) = \varnothing$.*

Together, the above two lemmas tell us that the right cosets of a subgroup $\mathbf{H}$ form a partition of the group $\mathbf{G}$, as is illustrated in the more accurate illustration below.



Using additional group properties, we can see that each right coset of $\mathbf{H}$ look, as a set, just like $\mathbf{H}$.

Lemma 43. *If $b \in \mathbf{G}$, then $r_b : \mathbf{H} \to \mathbf{H} * b$ (defined as $r_b(x) := x * b$) is one-to-one and onto. In particular, if* **H** *is finite, then every right coset of* **H** *has the same number of elements as* **H**.

This leads us to a yet more accurate picture, which tells us exactly why the order of **H** should divide the order of **G**. If the number of right cosets of **H** in **G** is finite, we call this number the **index** of **H** in **G** and denote it by $[\mathbf{G} : \mathbf{H}]$.

H	**H**$*b$	**H**$*c$	**H**$*d$	**H**$*f$	**H**$*g$	**H**$*k$

Lagrange's Theorem 44. *If* **G** *is a finite group and* **H** *is a subgroup of* **G**, *then* $\circ(\mathbf{G}) = [\mathbf{G} : \mathbf{H}] \cdot \circ(\mathbf{H})$. *In particular,* $\circ(\mathbf{H}) | \circ(\mathbf{G})$, *and* $[\mathbf{G} : \mathbf{H}]$ *is also the number of left cosets of* **H** *in* **G**.

Corollary 45. *If* **G** *is a finite group and* $b \in \mathbf{G}$, *then* $\circ(b) | \circ(\mathbf{G})$.

Corollary 46. *If* **G** *is a finite group of order n and* $b \in \mathbf{G}$, *then* $b^n = e$.

We can now see why it was true that every 5 or 7 element group must be cyclic and, with no additional effort, why the same must be true for every group of prime order.

Theorem 47. *Every group of prime order is cyclic.*

In contrast to Theorem 47, we can use the idea of Problem 39 to exhibit non-cyclic groups of order n for many non-prime values of n. (Hint: Show that if $n = p^2q$ where p is a prime, then $\mathbf{Z}_p \times \mathbf{Z}_{pq}$ is not cyclic.)

Theorem 48. *If every group of order $n > 1$ is cyclic, then n is either prime or a product of distinct primes.*

Problem 49. *Partition the numbers from 1 to 60 into three sets as follows.*

$A := \{n \mid$ *You can prove that every group of order n is cyclic.*$\}$,

$B := \{n \mid$ *You can exhibit a group of order n that is not cyclic.*$\}$

$C := \{n \mid$ *Neither of the above is true.*$\}$